

SHANA YUNSHENG LI

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EDUCATION

University of Illinois Urbana-Champaign	2024 - Now
Ph.D. in Mathematics	Advisor: Nathan Dunfield
Southern University of Science and Technology	2022 - 2024
M.Sc in Mathematics	Advisor: Stavros Garoufalidis
Southern University of Science and Technology	2018 - 2022
B.Sc in Mathematics and Applied Mathematics with Honors	

RESEARCH INTEREST

Low-dimensional Topology; Mathematical Physics; Quantum Topology; Knot Theory; Hyperbolic 3-Manifolds.

ACADEMIC VISITS

Simons Laufer Mathematical Sciences Institute	Jan. 2026 - Feb. 2026
Program Associate	Topological and Geometric Structures in Low Dimensions

GRANTS & FUNDING

ACCESS Discover Allocation (MTH260046, 1,500,000 ACCESS Credits)	2026 - 2027
Principal Investigator	The 12-tetrahedra census of orientable cusped hyperbolic 3-manifolds

PUBLICATIONS

5. (with Stavros Garoufalidis and Josephine Yu) Positivity and concavity of the colored Links–Gould polynomials of knots, 17 pages, arXiv:2605.30754.
4. Fixed-parameter tractable computation of Reshetikhin–Turaev knot polynomials via tensor networks, 13 pages, arXiv:2606.29662.
3. The complete 10-tetrahedra census of orientable cusped hyperbolic 3-manifolds, Proceedings of the 42nd International Symposium on Computational Geometry (SoCG 2026), Volume 367, pages 73:1-73:15. Full version: arXiv:2512.02142.
2. (with Stavros Garoufalidis) Patterns of the V_2 -polynomial of knots, Experimental Mathematics (2026), 1-31. arXiv:2409.03557.
1. (with Ni An and Stavros Garoufalidis) Algebraic aspects of holomorphic quantum modular forms, Research in Mathematical Sciences. 11 (2024) no. 3, Paper No. 54, 1-20.

PROJECTS

4. (with Stavros Garoufalidis) The loop expansion of colored knot polynomials for Lie superalgebras. Ongoing
3. Extending tabulation of quasi-alternating links. Ongoing
2. Census of 12 tetrahedra oriented cusped hyperbolic 3-manifolds. Ongoing
1. Census of 11 tetrahedra oriented cusped hyperbolic 3-manifolds. Drafting

THESES

M.Sc	On the Quantum Modularity Conjecture for Knots
B.Sc	Foundation of Supergeometry and Its Application in Mathematical Physics

TALKS

May. 2026	Invited online talk at Graduate Topology Seminar (SUSTech). Topic: Introduction to Reshetikhin–Turaev knot polynomials.
Mar. 2026	Seminar talk at Graduate Geometry-Topology Seminar (UIUC). Topic: The complete census of orientable cusped hyperbolic 3-manifolds, up to 11 tetrahedra.
Feb. 2026	Invited seminar talk at Computation in Geometric Topology, Topological and Geometric Structures in Low Dimensions, at Simons Laufer Mathematical Sciences Institute (SLMath). Topic: The complete census of orientable cusped hyperbolic 3-manifolds, up to 11 tetrahedra.
Nov. 2025	Invited online talk at Australian Geometric Topology Webinar. Topic: Multivariable knot polynomials, the V_n -polynomials, and their patterns.
Oct. 2025	Invited talk at AMS 2025 Fall Central Sectional Meeting at Saint Louis University, Special Session on Foliations and 3-manifolds. Topic: The complete census of orientable cusped hyperbolic 3-manifolds, up to 10 tetrahedra.
Sep. 2025	Online talk at Equivariant Topological Quantum Field Theory workshop at Casa Matemática Oaxaca (CMO). Topic: Multivariable knot polynomials, the V_n -polynomials, and their patterns.
Sep. 2025	Seminar talk at Group, Geometry and Topology Seminar (UIUC). Topic: Multivariable knot polynomials, the V_n -polynomials, and their patterns.
May. 2025	Conference talk at GTA Philadelphia 2025, 10th annual Graduate Student Conference in Algebra, Geometry, and Topology (GSCAGT) at Temple University Topic: Multivariable knot polynomials, the V_n -polynomials, and their patterns.
Apr. 2025	Invited online talk at Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS). Topic: Multivariable knot polynomials, the V_n -polynomials, and their patterns.
Mar. 2025	Seminar talk at Graduate Geometry-Topology Seminar (UIUC). Topic: Multivariable knot polynomials, the V_n -polynomials, and their patterns.
Mar. 2024	Seminar talk at Graduate Topology Seminar (SUSTech). Topic: The Teichmüller TQFT and the quantum modularity conjecture.

TEACHING

Spring 2025	IML graduate leader of undergraduate research project Visualizing Spaces of Curves in 3-manifolds.
Autumn 2022	Teaching assistant of MA109 (Advanced Linear Algebra).

AWARDS

Spring 2024	Outstanding Graduate Graduates Award Outstanding Master Thesis Award
Autumn 2022	Outstanding Teaching Assistant Award
Spring 2022	Outstanding Undergraduate Graduates Award Outstanding Undergraduate Thesis Award

OUTREACH - PIANO PERFORMANCES

Autumn 2023	“Moment · Eternity” Concert	Steinway Garden in Shenzhen Concert Hall
Spring 2022	“Youth Poetry and Music” Chamber Music Concert	Shenzhen Concert Hall
Autumn 2021	National Science & Technology University Aesthetic Education Seminar Concert	Steinway Garden in Shenzhen Concert Hall
Autumn 2020	Lang Lang’s Piano Masterclass	SUSTech 10th Anniversary
Autumn 2019	Haochen Zhang’s Piano Masterclass	